Royal Devon and Exeter NHS
NHS Foundation Trust

Clinical Guideline for:
Temperature control in patients who have suffered an out of hospital cardiac arrest

SUMMARY

2 small RCTs[1 2] published in 2002 demonstrated hypothermia for 24 hrs following out of hospital VF arrest led to a better neurological outcome. However, a recent much larger RCT [3] has demonstrated no benefit in cooling to 33°C over maintaining temperature at 36°C.

This guideline incorporates this into the care of the patient who suffers a cardiac arrest out of hospital, presumed to be secondary to a cardiac cause, and irrespective of arresting rhythm.

KEY POINTS

- Post cardiac arrest patients no longer need to be cooled to <34°C
- Maintaining temperature at 36°C is the aim initially, thereafter avoiding pyrexia >37°C
- Temperature control should follow step wise approach
- Not all patient require active cooling
- Sedation is required for the first 24hrs whilst pyrexia is being actively managed

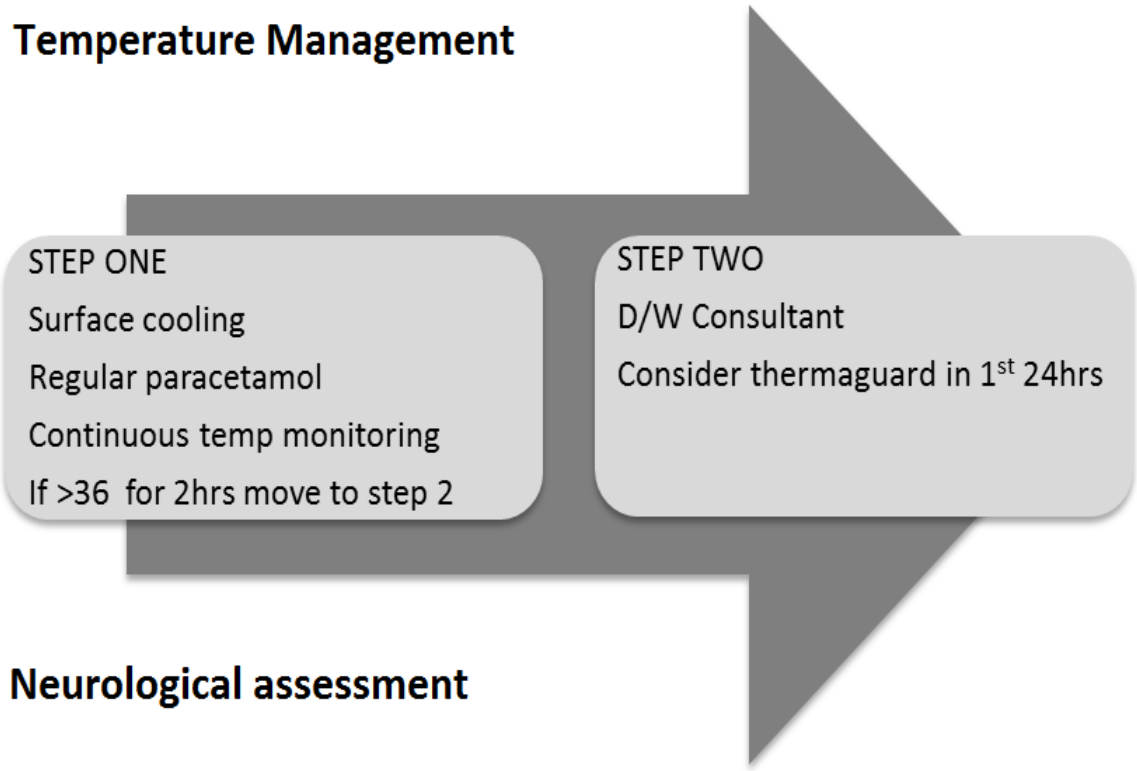
Exclusions

- Inpatients at the time of cardiac arrest
- Conscious patients obeying verbal commands, post ROSC
- Cardiac arrest secondary to non-cardiac cause
- Unwitnessed asystole
- Temp <30°C on presentation
- BP<80mmHg despite fluid loading, inotropes/balloon pump

Timeline

PHASE 1 – 36 hours post admission to ICU	PHASE 2 – next 36 hours	PHASE 3 - 36 hours after end of phase 2
• Maintain 36°C • Sedate & Ventilate • CT head/neck • PCI • EEG/SSEP if seizures • Start warming to 37°C at 28 hours	• Keep temperature <37°C until end of phase 2 • Sedation off at 36 hours post presentation • Assess for extubation, if extubated strict temp control no longer required • Revert to normal temperature management at end phase 2	• Multimodal neurological assessment at end of phase 3 (108 hours post presentation) to include EEG, SSEPs and clinical assessment on patients not regaining consciousness

Standard supportive intensive care including 'FASTHUG' approach



Findings allowing for consideration of withdrawal of active intensive care:

1) Brain dead due to cerebral herniation.

2) Severe myoclonus status * in the first 24 hours after admission and a bilateral absence of N20-peak on median nerve SSEP.

3) Ethical reasons**

4) After 108 hours post presentation: persisting coma with a Glasgow Motor Score 1-2 and bilateral absence of N20-peak on median nerve SSEP.

5) After 108 hours post presentation: persisting coma with a Glasgow Motor Score 1-2 and a treatment refractory status epilepticus

* diagnosed on EEG.

** such as further history of poor comorbid state, advanced directive

REFERENCES

1. Bernard SA, Gray TW, Buist MD, et al. Treatment of Comatose Survivors of Out-of-Hospital Cardiac Arrest with Induced Hypothermia. *N Engl J Med* 2002;**346**:557–63

2. HACA Study Group. Mild Therapeutic Hypothermia to Improve the Neurologic Outcome after Cardiac Arrest. *N Engl J Med* 2002;**346**:549–56

3. Nielsen N, Wetterslev J, Cronberg T, et al. Targeted Temperature Management at 33°C versus 36°C after Cardiac Arrest. *N Engl J Med* 2013;**369**:2197–206

PUBLICATION DETAILS

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